

Trainings ▾

Preparatory Steps

Jacket

WARNING

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To avoid arcing, remove the negative (-) battery cable first, and attach the negative (-) battery last.

WARNING

Do not remove the pressure cap from a hot engine. Wait until the coolant temperature is below 50°C [120°F] before removing the pressure cap. Heated coolant spray or steam can cause personal injury.

WARNING

Coolant is toxic. Keep away from children and pets. If not reused, dispose of in accordance with local environmental regulations.

- Disconnect the batteries or air supply line to the air starter to prevent accidental engine starting. See equipment manufacturer service information.
- Drain the cooling system. Refer to Procedure 008-018 in Section 8.
(/qs3/pubsys2/xml/en/procedures/56/56-008-018-tr.html)

Dual Impeller

WARNING

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To avoid arcing, remove the negative (-) battery cable first, and attach the negative (-) battery cable last.

WARNING

Coolant is toxic. Keep away from children and pets. If not reused, dispose of in accordance with local environmental regulations.

Note : This procedure is applicable to engines with dual impeller water pumps only. For engines with a single impeller water pump, reference the Low Temperature Aftercooler Water Pump procedure and the Jacket Water Pump Section in this procedure.

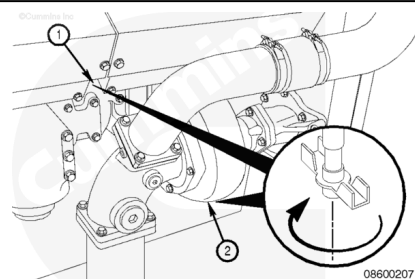
- Disconnect the battery cables. See equipment manufacturer service information.
- Drain the cooling system. Refer to Procedure 008-018 in Section 8.
(/qs3/pubsys2/xml/en/procedures/56/56-008-018-tr.html)
- Remove the alternator drive belt. Refer to Procedure 013-005 in Section 13.
(/qs3/pubsys2/xml/en/procedures/56/56-013-005-tr.html)
- Remove the alternator drive pulley. Refer to Procedure 009-010 in Section 9.
(/qs3/pubsys2/xml/en/procedures/56/56-009-010-tr.html)

Remove

Jacket

Note : The procedure for the QSK45 engine is shown. The procedure for the QSK60 engine is similar.

Open the draincock on the engine (1) and on the water pump (2).

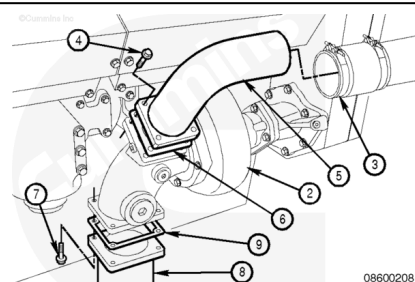


Remove the bypass tube hose clamp (3).

Remove the four capscrews (4), the bypass tube (5), and the gasket (6).

Remove the four capscrews (7), the inlet tube (8), and the gasket (9) from the water pump inlet.

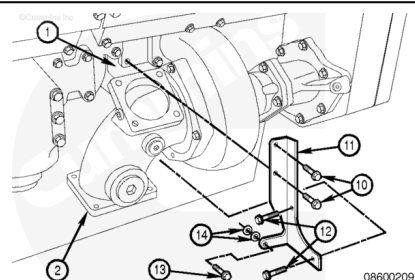
Discard the gaskets.



Remove the two capscrews (10) from the engine (1) and the water pump support bracket (11).

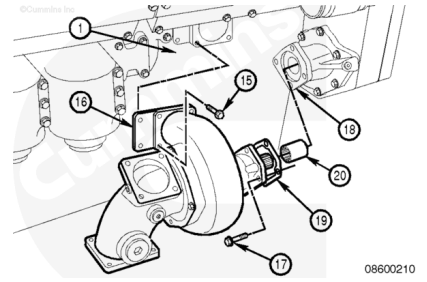
Remove the two capscrews (12) from the water pump (2) and the water pump support bracket (11).

Remove the capscrew (13), two spacers (14), and the water pump support bracket (11) from the water pump (2), and the engine (1).



⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.



Remove the four capscrews (15) from the water pump adapter (16) and the engine (1).

Remove the four capscrews (17) from the water pump and the water pump drive assembly (18).

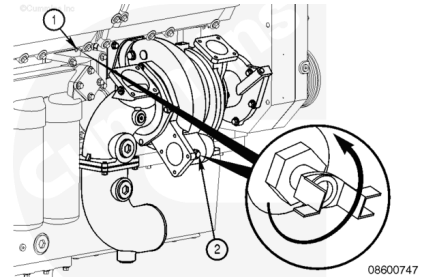
Remove the water pump, gasket (19), and the collar (20) from the water pump drive assembly (18).

Discard the gasket.

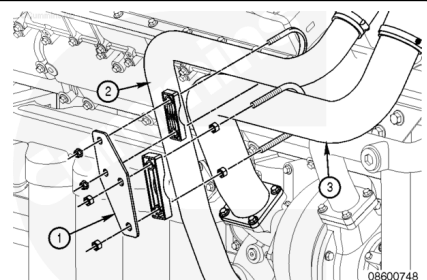
Dual Impeller

Note : The procedure for the QSK60 Tier II engine is shown. The procedure for the QSK50 Tier II is similar.

Open the draincock on the engine block (1) and the water pump (2).



Remove the two brackets and U-clamps (1) from the low temperature aftercooler outlet (2) and jacket water bypass pipes (3).

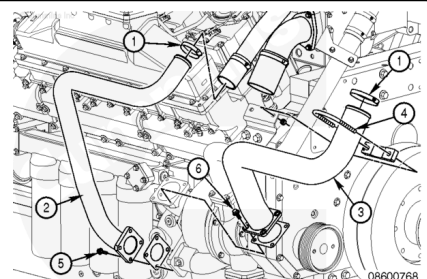


Remove the hose clamps (1) from the low temperature aftercooler outlet (2) and the jacket water bypass pipe (3).

Remove the U-clamp (4) from the jacket water bypass pipe (3).

Remove the four capscrews (5), low temperature aftercooler outlet (2), and the gaskets.

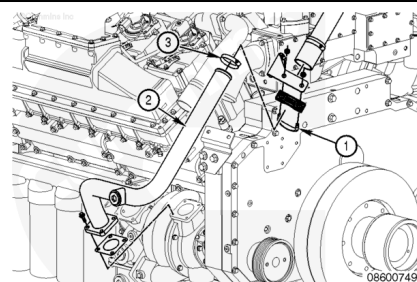
Remove the four capscrews (6), the jacket water bypass pipe (3), and the gaskets.



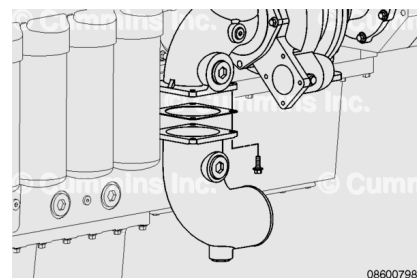
Remove the bracket/clamp (1) from the low temperature aftercooler bypass pipe (2).

Remove the hose clamp (3) from the low temperature aftercooler bypass pipe (2).

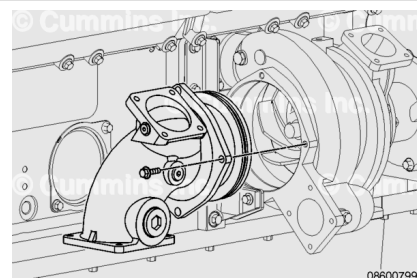
Remove the bypass pipe (2).



Remove the four capscrews and the jacket water radiator inlet connection.



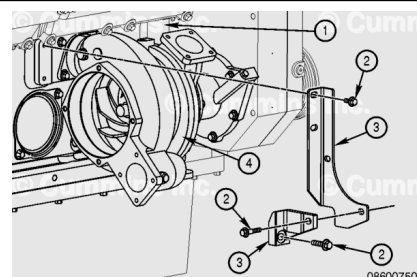
Remove the capscrews and the jacket water inlet connection.



Remove the two capscrews (2) from the engine block (1) and the water pump support bracket (3).

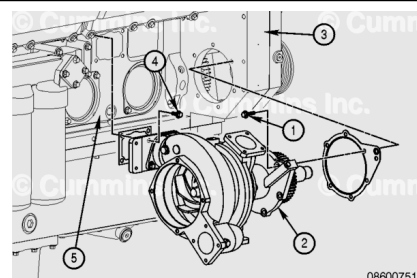
Remove the two capscrews (2) from the water pump (4) and the water pump support bracket (3).

Remove the capscrews (2) and the water pump support bracket (3) from the water pump (4) and the engine block (1).



⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.



Remove the six capscrews (1) from the water pump (2) and the gear housing (3).

Remove the four capscrews (4) from the engine block (5).

Remove the water pump from the front gear housing.

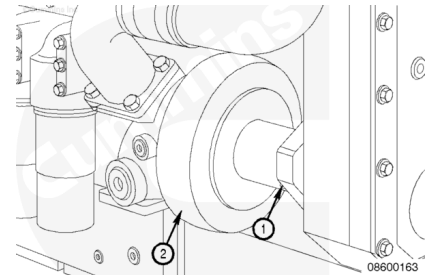
Inspect for Reuse

Jacket

A minor chemical buildup or streaking at the water pump weep hole (1) is normal. Do **not** repair or replace the water pump (2) unless an actual leak is confirmed. If a leak is confirmed, contact a Cummins® Authorized Repair Location for replacement.

Note : QSK45 and QSK60 engines have two-pump, two-loop systems. Check both pumps. Procedures for both pumps are similar.

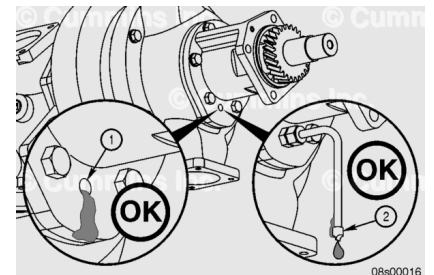
Inspect the water pump and bypass tube for coolant or oil leakage. Check the water pump weep hole for excessive leakage.



Dual Impeller

A minor chemical buildup or streaking at the water pump weep hole (1) is normal. Do **not** replace the water pump unless an actual leak is confirmed. If a leak is confirmed, contact a Cummins® Authorized Repair Location for replacement.

Note : Engines with a weep hole filter (2) installed may experience an occasional drip and minor chemical buildup from the filter. This is normal and does **not** indicate a leaking pump. Do **not** replace the water pump unless an actual leak is confirmed.



Disassemble

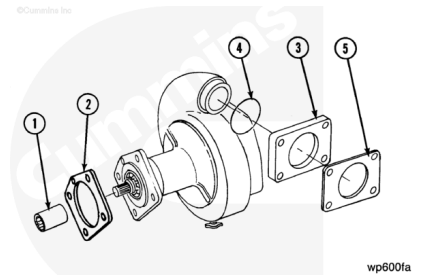
Jacket

Remove the:

- Splined coupling (1)
- Mounting gasket (2)
- Adapter plate (3)
- O-ring (4)

- Adapter plate gasket (5).

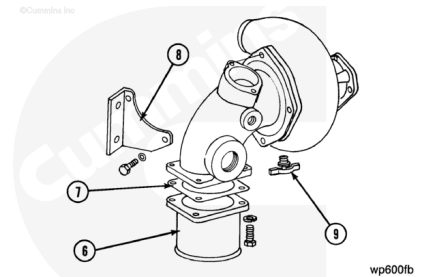
Discard the o-ring and gaskets.



Remove the:

- Water inlet connection (6)
- Gasket (7)
- Support bracket (8)
- Draincock (9).

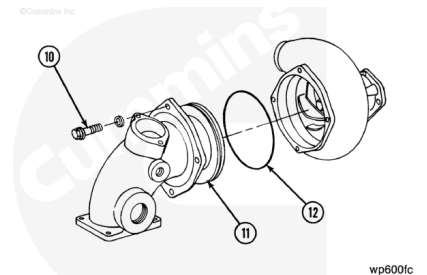
Discard the gasket.



Remove the:

- Capscrews (10)
- Water pump inlet housing (11)
- O-ring seal (12).

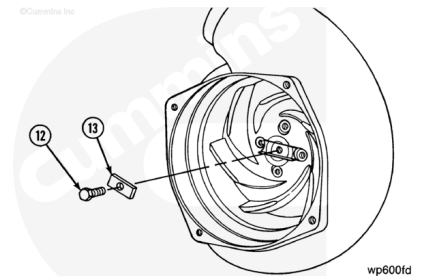
Discard the o-ring.



Note : New water pumps and rebuilt water pumps with new shafts and impellers do not contain the drive key and capscrew.

Remove the:

- Self-locking capscrews (12)
- Drive key (13).

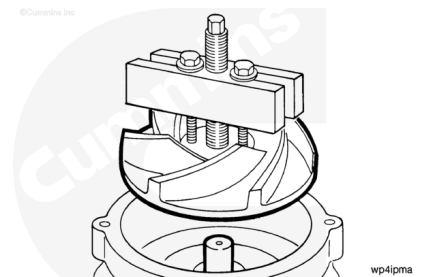


⚠ CAUTION ⚠

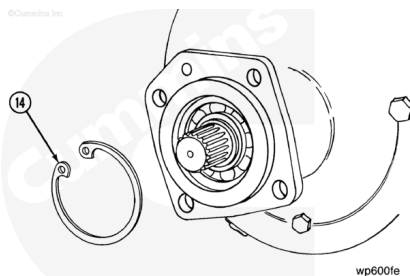
The puller screw must pass easily through the impeller bore to prevent damage to the impeller.

Install a short capscrew in the shaft to protect the threads.

Remove the impeller using a standard puller, Part Number ST-647, or equivalent.

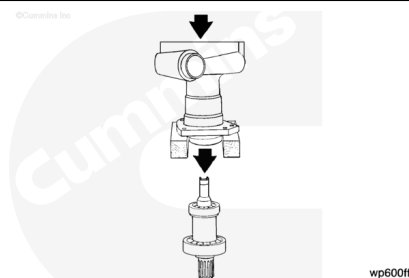


Remove the large snap ring (14).



Set the water pump in an arbor press with the impeller end of the shaft up.

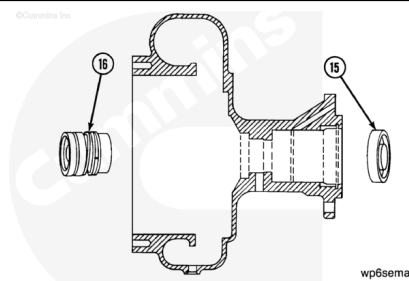
Press the bearing and shaft assembly out of the water pump housing.



Remove the:

- Oil seal (15)
- Mechanical water seal (16).

Discard the seals.



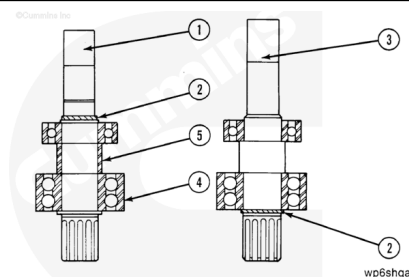
The bearing and shaft assembly can be one of two different designs. The disassembly and assembly procedures are different for the two designs.

The new design bearing and shaft assembly (1) positions the snap ring (2) nearest the impeller end of the shaft.

The previous design bearing and shaft assembly (3) positions the snap ring (2) nearest the end of the shaft.

The new design bearing shaft assembly can have either a double or single row bearing (4). If a single row bearing is used, a longer spacer (5) is required.

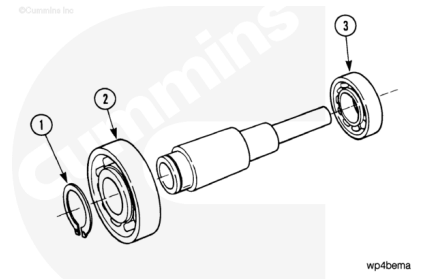
The previous design bearing and shaft assembly **must** use the double row bearing.



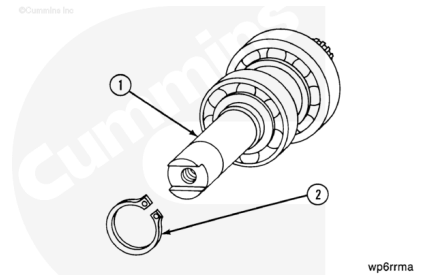
The following disassembly procedure is for the previous design.

Remove the snap ring (1).

Use water pump bearing separator, Part Number 3375326, or equivalent, to remove bearings (2) and (3) from the shaft.

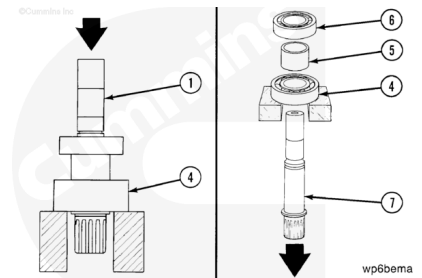


Remove the snap ring (2).



Place the bearing and shaft assembly (1) onto supports with the large bearing (4) on the supports.

Use a hydraulic press to press the shaft (7) through the bearing (4), spacer (5), and bearing (6).



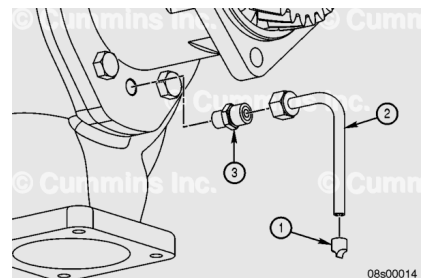
Dual Impeller

Note : There are no replacement parts or disassembly processes for the dual impeller water pump assembly.

Remove the filter (1) from the water pump weep hole drain tube.

Remove the drain tube (2) from the weep hole fitting (3).

Remove the fitting (3) from the water pump.



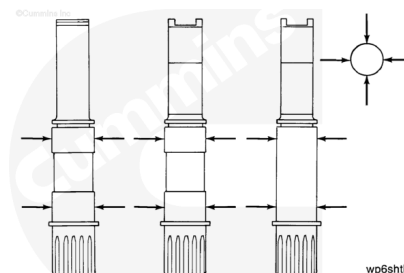
Measure

Jacket

Measure the water pump shaft bearing area diameter.

Water Pump Shaft Diameter (bearing area)

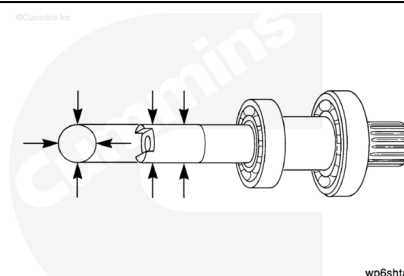
mm		in
30.010	MIN	1.181
30.020	MAX	1.812



Measure the diameter of the impeller end of the water pump shaft.

Water Pump Shaft Diameter (impeller end)

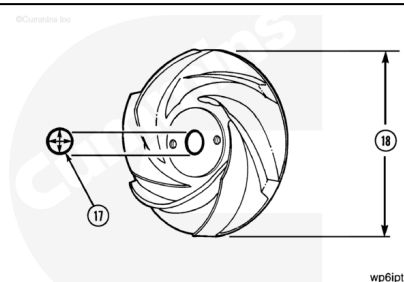
mm		in
24.999	MIN	0.984
25.009	MAX	0.985



Measure the inside diameter (17) of the water pump impeller.

Water Pump Impeller Inside Diameter, Part Numbers 3172018 and 3172019

mm		in
24.919	MIN	0.981
24.945	MAX	0.982



Water Pump Impeller Inside Diameter, Part Numbers 5611531 and 5611532

mm		in
24.563	MIN	0.967
24.589	MAX	0.968

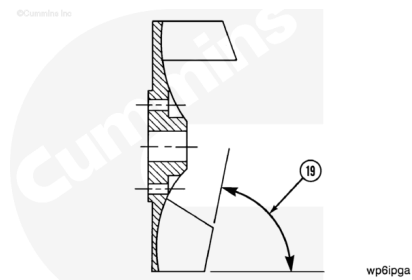
The outside diameter (18) of the water pump impeller is given for parts identification **only**. It is **not** necessary to measure the part.

Water Pump Impeller Outside Diameter

Engine	mm	in
QSK45	146	5.75
QSK60	171	6.75

Water pump impellers with a 74 degree (19) angle can **only** be installed with a water pump connection that has a 16 degree angle.

Water pump impellers with a 67 degree angle (19) can **only** be installed with a water pump connection that has a 23 degree angle.

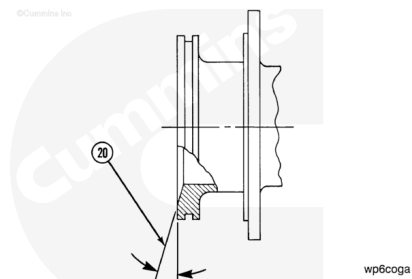


QSK45 Water Pump Impeller Angle

Part Number	Angle
3050455	74 degrees
3175069	74 degrees
3050453	67 degrees
3001711	67 degrees

All QSK60 impellers have a 78 degree angle (19).

The water pump inlet housing angle (20) is measured at the location illustrated.

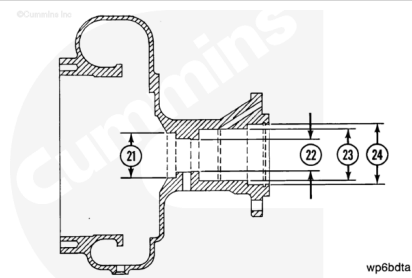


QSK45 Water Pump Inlet Housing Angle

Part Number	Angle
2006456	16 degrees
3001712	23 degrees

All QSK60 water pump inlet housings have a 12 degree angle (20).

Measure the water pump housing mechanical water seal bore (21), oil seal bore (22), rear bearing bore (23), and front bearing bore (24).



Water Pump Housing Mechanical Water Seal Bore

mm		in
53.60	MIN	2.110
54.10	MAX	2.130

Water Pump Housing Oil Seal Bore

mm		in
38.05	MIN	1.498
38.10	MAX	1.500

Water Pump Housing Rear Bearing Bore

mm		in
61.997	MIN	2.440
62.009	MAX	2.441

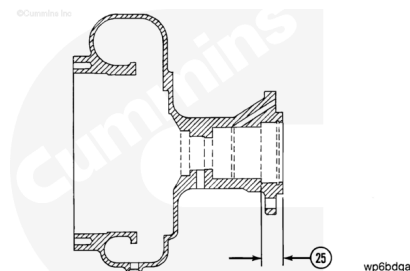
Water Pump Housing Front Bearing Bore

mm		in
71.996	MIN	2.834
72.012	MAX	2.835

The depth of the water pump housing front bearing bore (25) is different for single row bearings and double row bearings.

Water Pump Housing Front Bearing Bore Depth

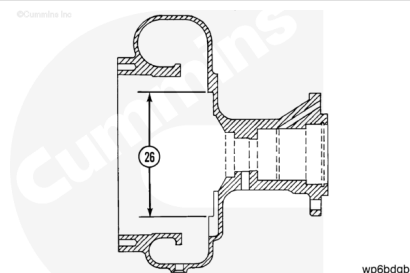
Bearing Style	mm	in
Single row bearing	25.4	1.00
Double row bearing	36.8	1.45



The water pump housing impeller counterbore diameter (26) is different between engines.

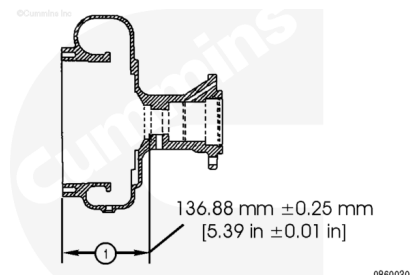
Water Pump Housing Impeller Counterbore Diameter

Engine	mm	in
QSK45 (new)	147	5.80
QSK45 (previous)	160	6.30
QSK60	173	6.80



⚠ CAUTION ⚠

Do not install a new mechanical water seal in an existing water pump housing without machining the pump housing. Doing so can cause rapid mechanical water seal failure.



Assemble

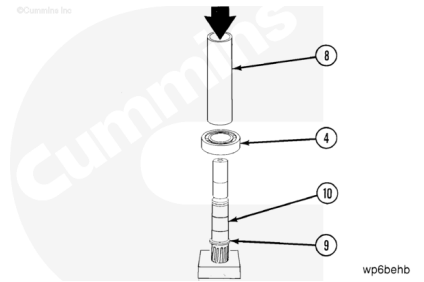
Jacket

⚠ CAUTION ⚠

Make sure to install the correct bearing. If the wrong bearing is installed, the pump will fail.

⚠ CAUTION ⚠

To prevent damage to the bearing, the mandrel must contact the inner race of the bearing during installation.



Note : The following three steps are for the new shaft assembly.

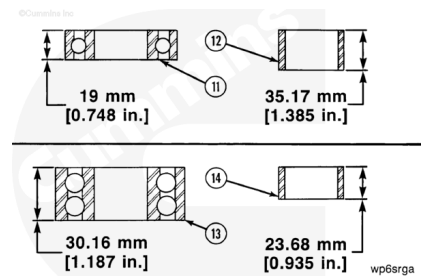
Support the shaft on the splined end as shown in the illustration.

Use a hydraulic press and the appropriate mandrel (8) to press the bearing onto the shaft until it rests against the step on the shaft (9).

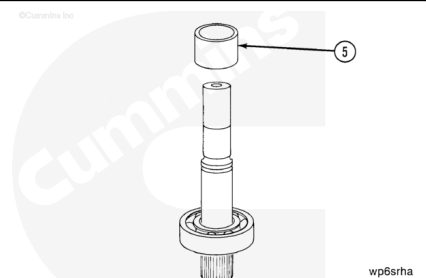
On new shafts, the bearing will **not** fit tightly in the area of the shaft (10) that has been machined to a smaller diameter.

If a single row bearing is used (11), the long spacer (12) **must** be used.

If a double row bearing is used (13), the short spacer (14) **must** be used.



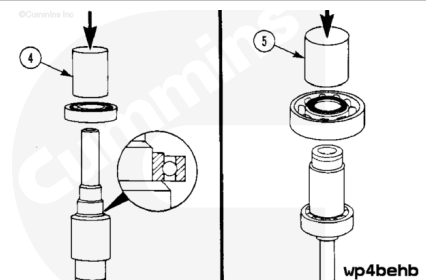
Slide the correct spacer (5) onto the shaft until it touches the inner race of the bearing.



⚠ CAUTION ⚠

To prevent damage to the bearing, the mandrel must contact the inner race of the bearing during installation.

Note : The following bearing and shaft assembly steps are for the previous design. The remainder of the assembly procedures are the same for the new and previous designs.

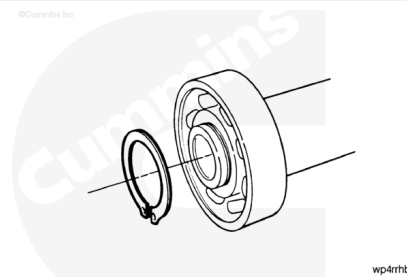


Only double row bearings can be used for the front bearing on the previous design.

Support the shaft as shown in the illustration.

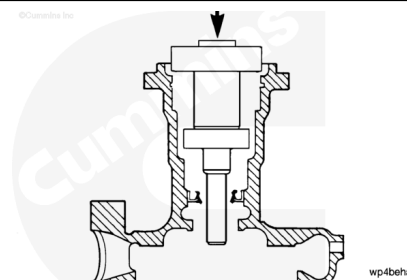
Use an arbor press and the appropriate mandrel (4) and (5) to press each bearing onto the shaft until it rests against the step on the shaft.

Install the snap ring.



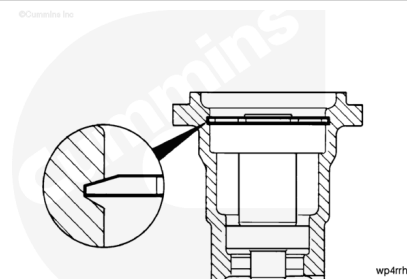
Support the water pump housing on an arbor press.

Use the arbor press and water pump bearing driver, Part Number 3375318, or equivalent, to press the bearing and shaft assembly into the water pump housing.



The beveled edge of the snap ring **must** be positioned up as shown in the illustration.

Install the snap ring.

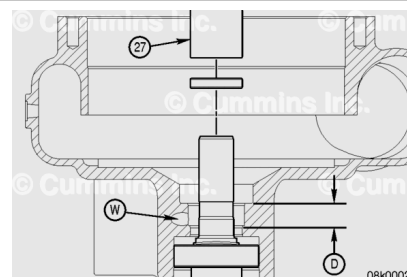


⚠ CAUTION ⚠

When installing the seal, take care not to damage the seal lip.

Press the oil seal onto the shaft using an oil seal mandrel. If an oil seal mandrel is **not** available, press the seal onto the shaft backward to open up the seal lip, then turn the seal the correct way around.

Use the mechanical water pump seal installer (27), to press the water pump seal into the water pump housing past the weep hole cavity.



Reference the Tool Catalog, Publication Number 3377710 to determine which tool number to use based on the water pump seal part number.

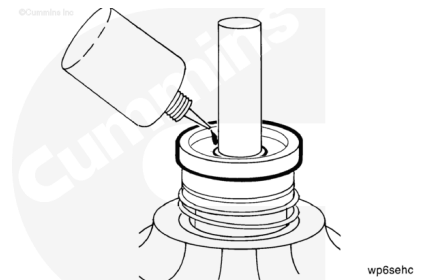
The installed distance (D) between the oil seal face from the impeller side and the inner face in the impeller housing **must** be 15.60 to 16.79 mm [0.614 to 0.661 in].

Check after the installation that the oil seal is **not** obstructing the weep hole (W) entry.

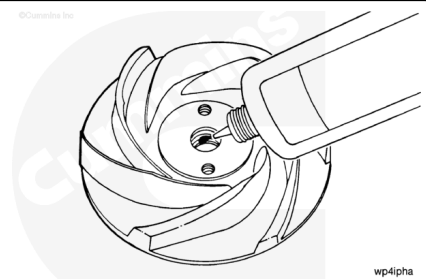
⚠ CAUTION ⚠

The mechanical water seal will fail if the spring tension is not correct. Use Loctite™ to keep the seal in place. If Loctite™ contacts the sealing surfaces, the seal will fail.

Apply **only** one drop of Loctite™ 290, or equivalent, to the seal at the parting line of the shaft.



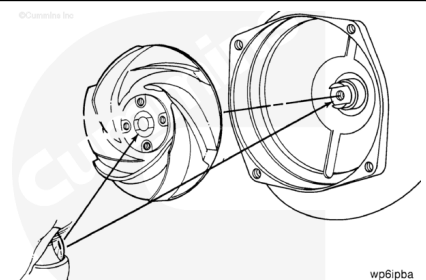
Apply a light coat of Loctite™ 609, or equivalent, to the inside area of the impeller shaft bore.



⚠ CAUTION ⚠

If the incorrect impeller or water pump inlet connection is installed, the water pump will be damaged from contact, or the performance of the pump will be decreased.

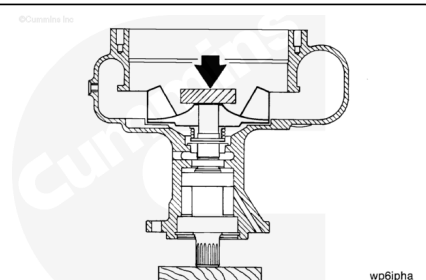
For water pumps that have impeller and shaft slots for a drive key, the slots **must** be aligned before the impeller is pressed onto the shaft.



Support the water pump on the shaft as shown in the illustration.

Place a small piece of flat steel on the impeller.

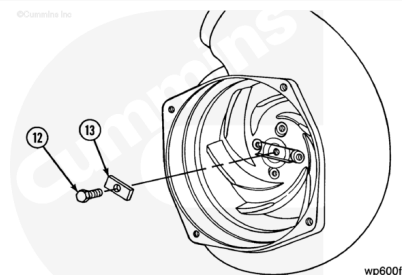
Use an arbor press to press the impeller onto the shaft until it is even with the end of the shaft.



For impellers that have slots for a drive key, install the drive key (13) and self-locking capscrews (12).

Tighten the capscrews.

Torque Value: 20 n•m [177 in-lb]



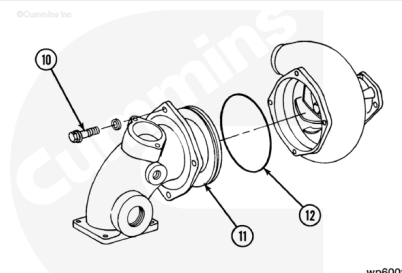
Lubricate the o-ring with vegetable oil.

Install the o-ring (12) onto the water pump inlet housing (11).

Install the water pump inlet housing into the water pump housing.

Install and tighten the capscrews.

Torque Value: 45 n•m [33 ft-lb]



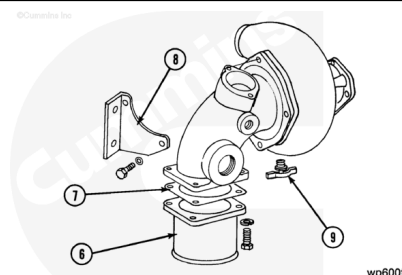
Install the gasket (7), water pump inlet connection (6), and capscrews.

Tighten the capscrews.

Torque Value: 40 n•m [30 ft-lb]

Install the draincock and tighten.

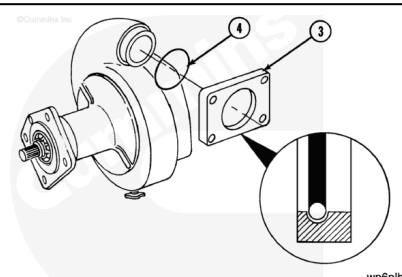
Torque Value: 15 n•m [133 in-lb]



Lubricate the o-ring (4) with vegetable oil.

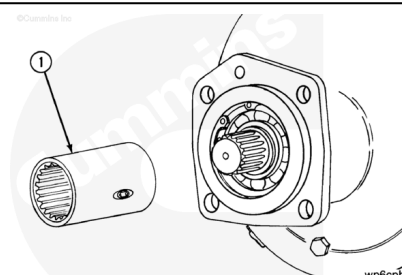
Install the o-ring in the groove of the adapter plate (3).

Install the adapter plate with the o-ring side toward the housing.



While rotating the shaft, lubricate the bearing with clean engine oil.

Lubricate the splines on the shaft and install the spline coupling.



Dual Impeller

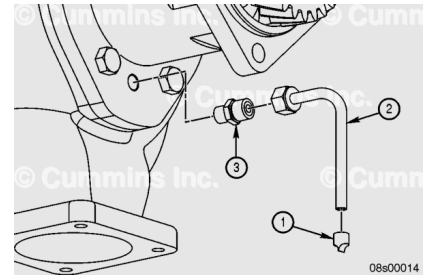
Install the weep hole drain tube fitting (3) onto the water pump and tighten.

Torque Value: 12 n•m [106 in-lb]

Install the weep hole drain tube (2) onto the fitting (3) and tighten the compression nut.

Torque Value: 12 n•m [106 in-lb]

Install the filter (1) onto the weep hole drain tube (2).



Install

Jacket

⚠ WARNING ⚠

This component weighs 23 kg [50 lb] or more. To reduce the possibility of personal injury, use a hoist or get assistance to lift this component.

Position the collar (20) on the water pump drive assembly (18).

The collar can be installed on the water pump shaft or the water pump drive assembly shaft. It is usually easier to install on the water pump drive assembly.

Install the water pump and gasket (19) on the water pump drive assembly (18) with four capscrews (17).

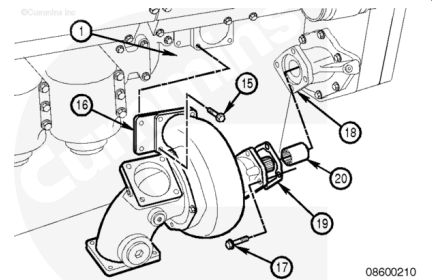
Tighten the capscrews.

Torque Value: 80 n•m [59 ft-lb]

Install the water pump adapter (16) on the engine (1) with four capscrews (15).

Tighten the capscrews.

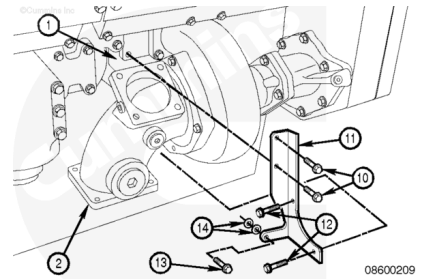
Torque Value: 45 n•m [33 ft-lb]



Install the water pump support bracket (11) on the water pump (2) and the engine (1) with two spacers (14), one screw (13), two screws (12), and two screws (10).

Tighten the capscrews.

Torque Value: 45 n•m [33 ft-lb]



Install the inlet tube (8) and gasket (9) on the water pump inlet with four capscrews (7).

Tighten the capscrews.

Torque Value: 45 n•m [33 ft-lb]

Install the bypass tube (5) and gasket (6) on the water pump inlet with four capscrews (4).

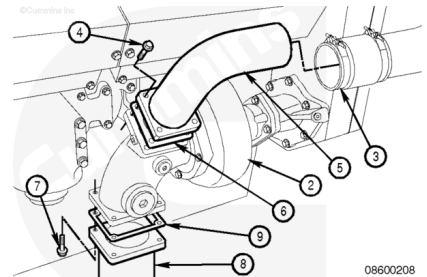
Tighten the capscrews.

Torque Value: 45 n•m [33 ft-lb]

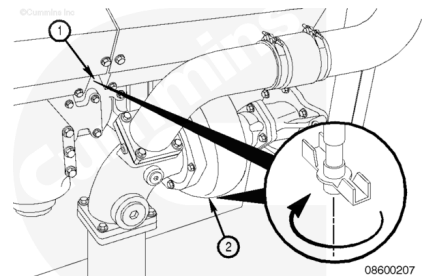
Install the bypass tube hose clamp (3).

Tighten the clamp.

Torque Value: 8 n•m [71 in-lb]



Close the draincock on the engine (1) and on the water pump (2).



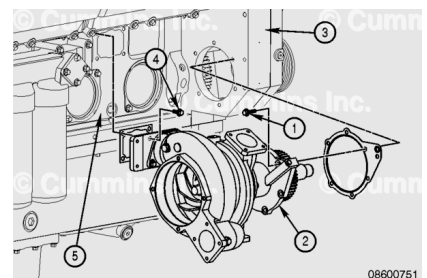
Dual Impeller

⚠ WARNING ⚠

This assembly weighs 23 kg [50 lb] or more. To reduce the possibility of personal injury, use a hoist or get assistance to lift this assembly.

Install the water pump (2) into the gear housing (3) using the six capscrews (1) and a new gasket.

Install the four capscrews (4) to the engine block (5) using a new gasket.



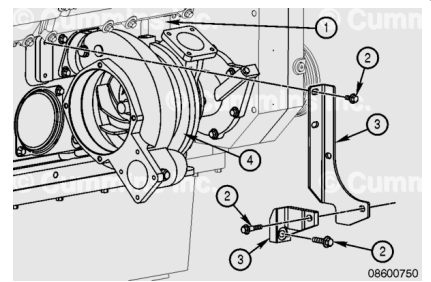
Tighten the capscrews.

Torque Value: 45 n•m [33 ft-lb]

Install the capscrews (2) and the water pump support bracket (3) to the water pump (4), and the engine block (1).

Install the two capscrews (2) to the water pump (4), and the water pump support bracket (3).

Install the two capscrews (2) to the engine block (1) and the water pump support bracket (3).

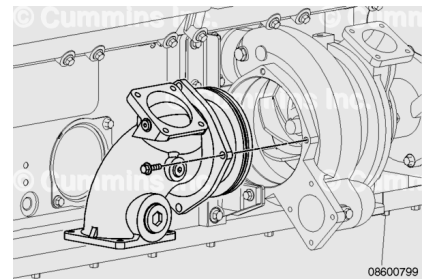


Torque Value: 45 n•m [33 ft-lb]

Lubricate the new o-ring on the jacket water inlet.

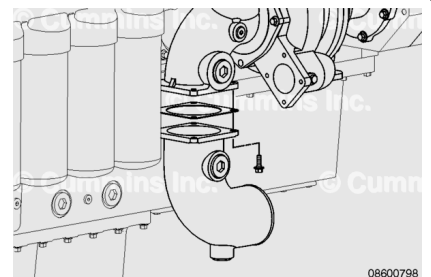
Install the jacket water inlet and capscrews onto the water pump.

Torque Value: 45 n•m [33 ft-lb]



Install the jacket water radiator inlet connection, gasket, and four capscrews.

Torque Value: 45 n•m [33 ft-lb]

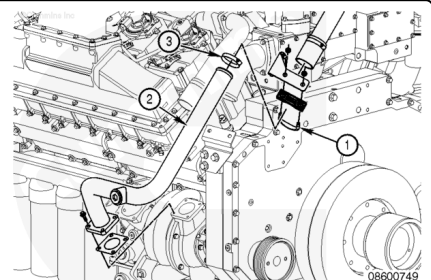


Install the bypass pipe (2).

Install the hose clamp (3) on the low temperature aftercooler bypass pipe (2).

Install the bracket/clamp (1) to the low temperature aftercooler bypass pipe (2).

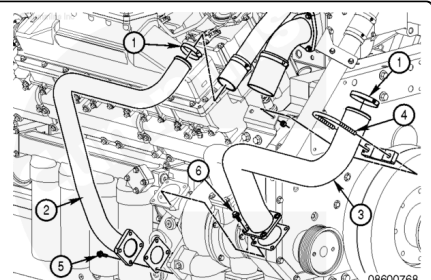
Torque Value: 45 n•m [33 ft-lb]



Install the jacket water bypass pipe (3), gaskets, and four capscrews (6).

Install the low temperature outlet pipe (2), gasket, and four capscrews (5).

Install the U-clamp (4) to the jacket water bypass pipe (3).



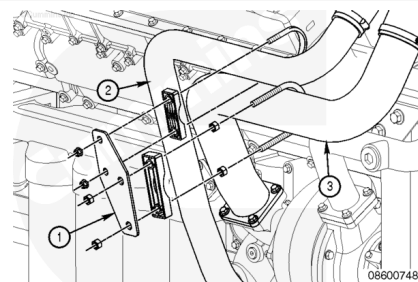
Install the hose clamps (1) to the low temperature aftercooler outlet (2) and the jacket water bypass pipe (3).

Torque Value: 45 n•m [33 ft-lb]

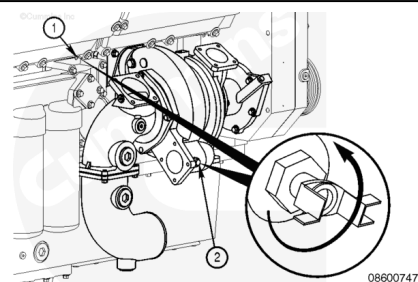
Install the bracket and U-clamps (1) to the low temperature aftercooler outlet (2) and jacket water bypass pipe (3)

Tighten the U clamps.

Torque Value: 45 n•m [33 ft-lb]



Close the draincock on the engine block (1) and on the water pump (2).



Finishing Steps

Jacket

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

⚠ WARNING ⚠

Coolant is toxic. Keep away from children and pets. If not reused, dispose of in accordance with local environmental regulations.

- Fill the cooling system. Refer to Procedure 008-018 in Section 8.
(/qs3/pubsys2/xml/en/procedures/56/56-008-018-tr.html)
- Connect the batteries. See equipment manufacturer service information.
- Operate the engine to 70°C [160°F] coolant temperature. Check for leaks.

Dual Impeller

WARNING

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

WARNING

Coolant is toxic. Keep away from children and pets. If not reused, dispose of in accordance with local environmental regulations.

- Connect the batteries. See equipment manufacturer service information.
- Fill the cooling system. Refer to Procedure 008-018 in Section 8.
(/qs3/pubsys2/xml/en/procedures/56/56-008-018-tr.html)
- Install the alternator drive pulley. Refer to Procedure 009-010 in Section 9.
(/qs3/pubsys2/xml/en/procedures/56/56-009-010-tr.html)
- Install the alternator drive belt. Refer to Procedure 013-005 in Section 13.
(/qs3/pubsys2/xml/en/procedures/56/56-013-005-tr.html)
- Operate the engine to 70°C [160°F] coolant temperature. Check for leaks.